

# Programming Using C++ Level 1

## Functions



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# Definition

- ◆ A function is a reusable block of code designed to perform a specific task.
- ◆ Functions help organize a program into smaller and more manageable parts.
- ◆ A function can be called whenever its task is needed.

## Explanation

◆ This program defines four functions, where each function performs a different task.

- `DisplayMyCardInfo()` → displays personal information.

- `PrintSquareStars()` → prints a square made of stars.

- `PrintILoveProgramming()` → prints a message.

- `PrintSLetter()` → prints the letter S using stars.

◆ The `main()` function calls these functions in sequence.

# ☐ Steps in main()

☐ Define the first function

◆ `DisplayMyCardInfo()` displays the card information.

☐ Define the second function

◆ `PrintSquareStars()` prints four lines of stars.

☐ Define the third function

◆ `PrintILoveProgramming()` prints a programming-related message.

☐ Define the fourth function

◆ `PrintSLetter()` prints the letter S using stars.

☐ Call the functions

◆ The `main()` function calls each function in the required order.

# 💡 Example

- ◆ A function is defined with:

```
void DisplayMyCardInfo()  
{  
    // Code  
}
```

- ◆ It is called using:

```
DisplayMyCardInfo();
```

## ➡ Result

- ◆ When the function is called, the code inside the function is executed.

## Note

- ◆ ``void`` means that the function does not return a value.
- ◆ The function name is followed by parentheses ``()``.
- ◆ The code that belongs to the function is placed inside ``{ }``.

## ⊘ Warning

- ◆ Defining a function does not execute it.
- ◆ The function must be called for its code to run.



# Comparison

## ▨ Function

- ✓ Contains code for a specific task.
- ✓ Can be called multiple times.
- ✓ Helps avoid repeating the same code.

## ▤ Without Functions

- ✓ Code is written directly inside ``main()`.`
- ✓ Repeated code may need to be written multiple times.
- ✓ Larger programs become harder to organize.

# ✦✦ Summary

- ✓ A function is a reusable block of code that performs a specific task.
- ✓ Functions are defined separately and executed when they are called.
- ✓ Using functions makes programs more organized and easier to maintain.

Thank you 😊